Experiment No: 3 **Experiment Name: Demonstrate missing value analysis, fixing missing values and outlier analysis in dataset**

Aim:

To identify and handle missing values in a dataset using various imputation techniques and to detect and treat outliers using statistical methods.

Algorithm:

Step 1: Import required libraries (pandas, numpy, matplotlib, seaborn).

Step 2: Load the dataset and inspect it using head(), info(), and describe().

Step 3: Missing Value Analysis:

a. Use isnull().sum() to count missing values per column.

b. Calculate the percentage of missing values.

Step 4: Fix Missing Values:

a. Drop rows/columns with excessive missing values.

b. Fill numerical columns with mean/median.

c. Fill categorical columns with mode.

Step 5: Outlier Analysis:

a. Use IQR (Interquartile Range) method to detect outliers.

b. IQR = Q3 - Q1; Outliers: values < Q1 - 1.5*IQR or > Q3 + 1.5*IQR.

Step 6: Treat Outliers by capping (Winsorizing) or removing them.

Step 7: Visualize using box plots before and after treatment.

Step 8: Display cleaned dataset summary.

Program:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Create sample dataset with missing values
np.random.seed(42)
data = pd.DataFrame({
    'Age':    [25, np.nan, 35, 28, np.nan, 42, 60, 31, 29, 150],
    'Salary': [50000, 60000, np.nan, 55000, 72000, np.nan, 80000, 45000,
58000, 62000],
    'Score':  [85, 90, 78, np.nan, 88, 76, 95, np.nan, 82, 91]
})

print("=== Original Data ===")
print(data)
```

```
# Missing Value Analysis
print("\n=== Missing Values ===")
print(data.isnull().sum())

# Fix Missing Values - fill with mean
data_clean = data.copy()
data_clean['Age'].fillna(data_clean['Age'].median(), inplace=True)
data_clean['Salary'].fillna(data_clean['Salary'].mean(), inplace=True)
data_clean['Score'].fillna(data_clean['Score'].mean(), inplace=True)

print("\n=== After Fixing Missing Values ===")
print(data_clean.isnull().sum())

# Outlier Analysis using IQR
def remove_outliers(df, col):
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    print(f"{col}: Lower={lower:.2f}, Upper={upper:.2f}")
    df[col] = np.clip(df[col], lower, upper)
    return df

print("\n=== Outlier Bounds ===")
for col in ['Age', 'Salary', 'Score']:
    data_clean = remove_outliers(data_clean, col)

# Boxplot
plt.figure(figsize=(10, 4))
data_clean.boxplot(column=['Age', 'Salary', 'Score'])
plt.title('Boxplot After Outlier Treatment')
plt.savefig('outlier_output.png')
plt.show()
print("\nData cleaned successfully.")
```

Output:

=== Missing Values ===

Age 2

Salary 2

Score 2

=== After Fixing Missing Values ===

Age 0

Salary 0

Score 0

=== Outlier Bounds ===
Age: Lower=13.50, Upper=51.50
Salary: Lower=33250.00, Upper=86750.00
Score: Lower=67.25, Upper=104.75

Data cleaned successfully.

Result:

Missing values were successfully identified and imputed using median/mean strategies. Outliers were detected using the IQR method and treated by capping. The dataset was cleaned and ready for further analysis.